ASSEMBLY INSTRUCTIONS

MICRODOCK

V3 · THE EGG DOCK

A 3D-printed charging dock for Microduck. It finds the tag, sits down and scoots back under the hood, which floats on printed springs hidden inside the egg. Charged, it scoots out and walks off.

6 PRINTED PARTS	**~9 h** PRINT, PLA
~15 min TO BUILD	**0** SCREWS OR GLUE



NTED SPRINGS ★ NO SCREWS ★ NO GLUE ★ ONE CABLE ★ MAGNETIC USB-C ★ FLOATING HOOD ★ EGG SHAPED ★ PRINTED SPRINGS ★ N

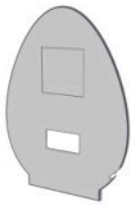
MicroDock v3 · for Microduck · rendered from the CAD and the simulation

WHAT YOU NEED

PRINT



1x Base



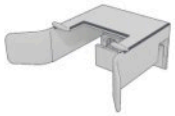
1x Front plate



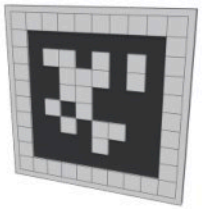
1x Dome



1x Spring stage



1x Hood



1x Tag



2x Filament pin, ~4 cm



Plate 1 base



Plate 2 front plate



Plate 3 dome



Plate 4 spring stage



Plate 5 hood



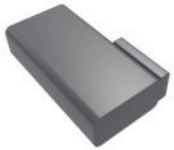
Plate 6 tag (two colours)

PLA, white, plus black for the tag. The Bambu Lab P1S project sets everything: six plates, about 9 h 15 min and 365 g (a seventh, optional plate prints a slice of the hood to try your adapters in first). The front plate and the hood print on supports, the dome with a brim. The pins are pieces of 1.75 mm filament.

GET



1x magnetic USB-C adapter (a pair): the adapter goes in the hood, the tip in Microduck

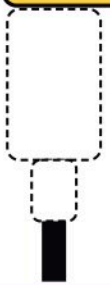


1x 90° USB-C adapter, "horizontal": it turns the cable sideways in the hood (goobay USB4 angle adapter, Kjell & Company 66960)



1x USB-C cable and charger: the hood holds this cable's plug (Clas Ohlson's 20 W kit, 39-2051: a white 60 W cable, 1 m)

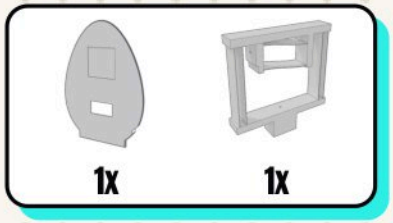
PLUG CHECK 1:1

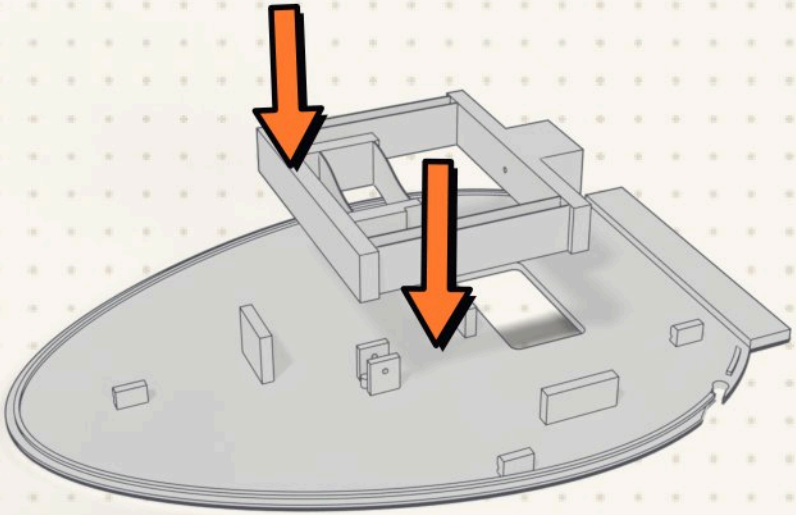


Print this page at 100 %. Your cable's plug should match the outline: 12.3 x 20 mm, 6.4 mm thick. The hood's clip is made for it.

ONLY A FLUSH CUTTER

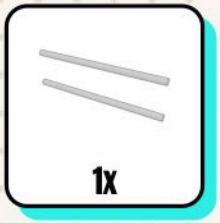
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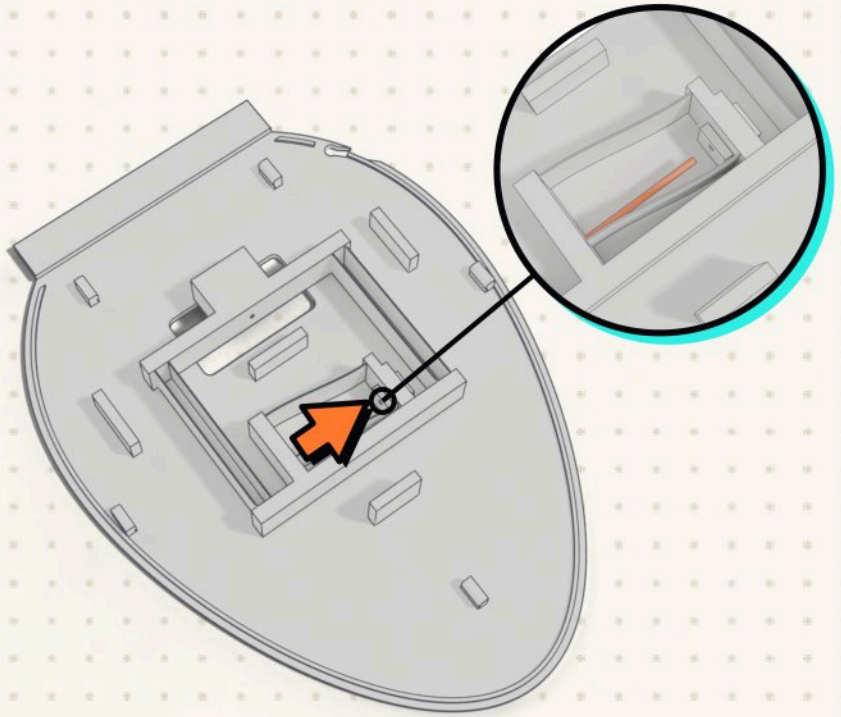




Lay the front plate face down. Put the spring stage on it: the post's two holes onto the plate's two pegs.

2



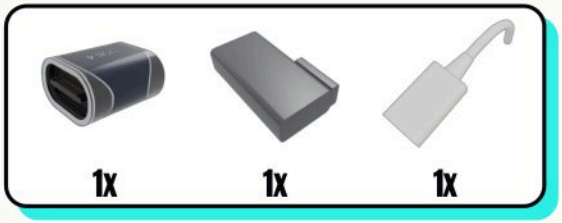


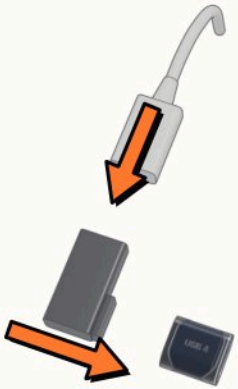
Push a pin through the ear, the post and the far ear, from the middle of the egg. Cut it off flush on both sides.

> The pin holds the post to the front plate: the push of every docking goes into the plate.

BUILD FIRST: HOOD + CABLE

3

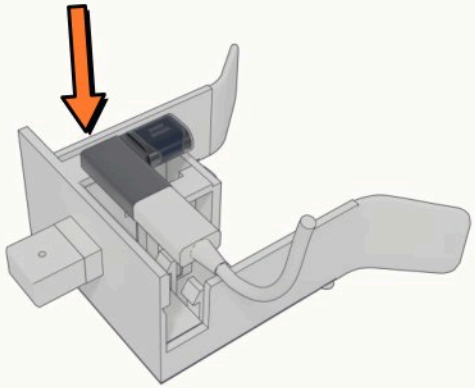




Plug the cable into the angle adapter's side, then the angle adapter into the magnetic adapter's back.

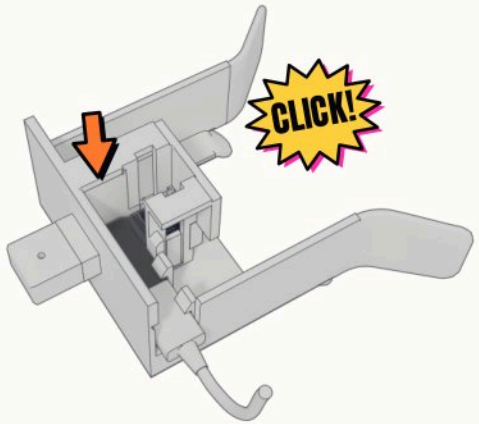
4





Turn the hood upside down. Lower the three in: the magnetic face toward the hood's flared end, the cable out through the opening in the side wall.

5

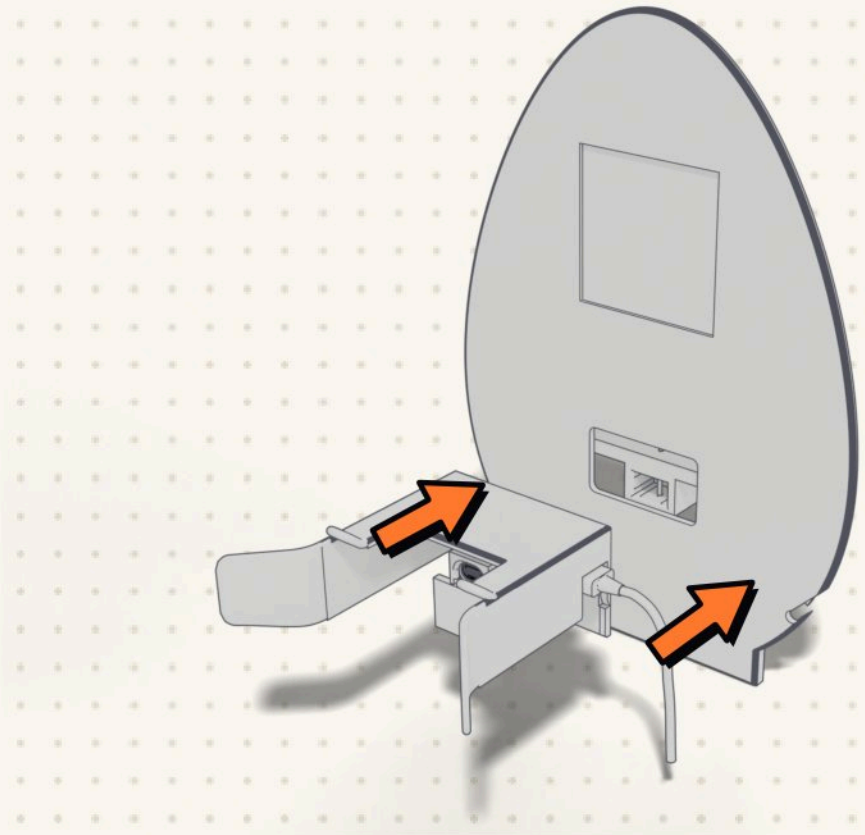


Hold it toward the flared end and press it down until it clicks.

> Two clips hold it: one under the angle adapter, one under the cable's plug. To take it out, push them toward the flared end and pull the three out.

6

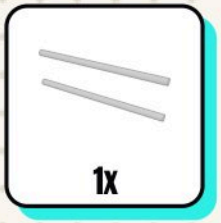


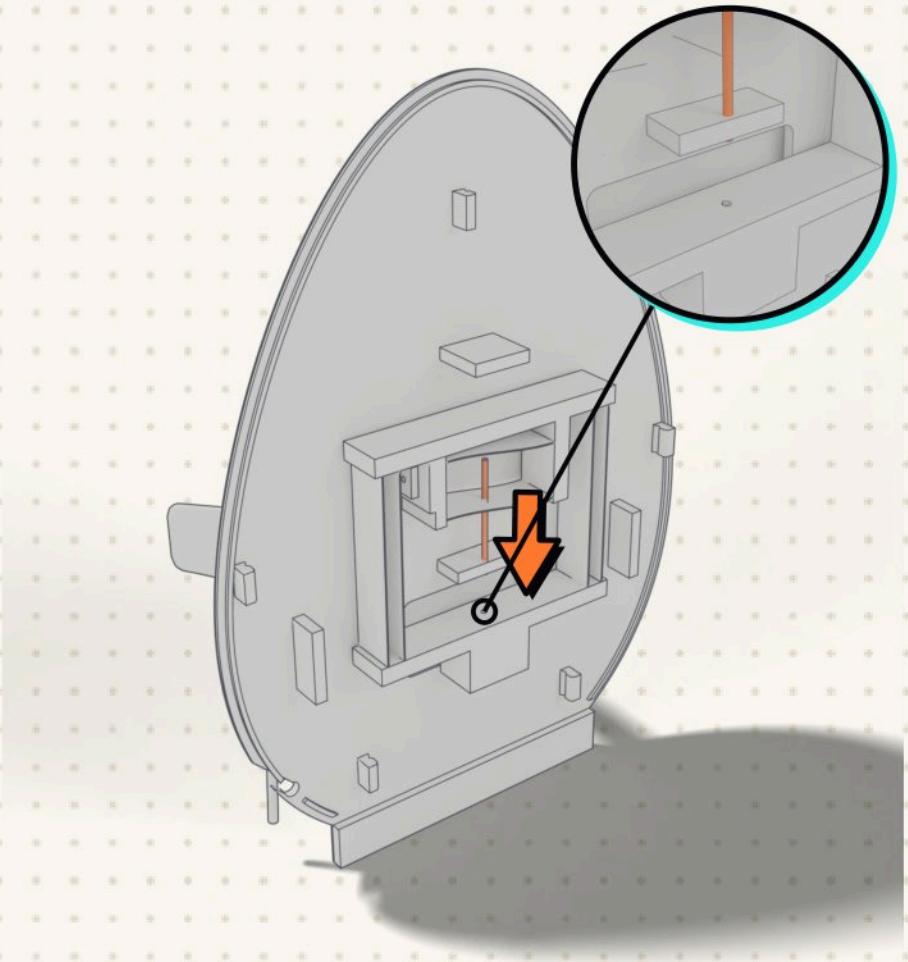


Turn the hood back over. Stand the front plate up and push the hood's arm through its window into the stage's carriage, all the way in. Hold the carriage from behind.

Crush ribs in the carriage hold the arm with no play: it takes a firm push.

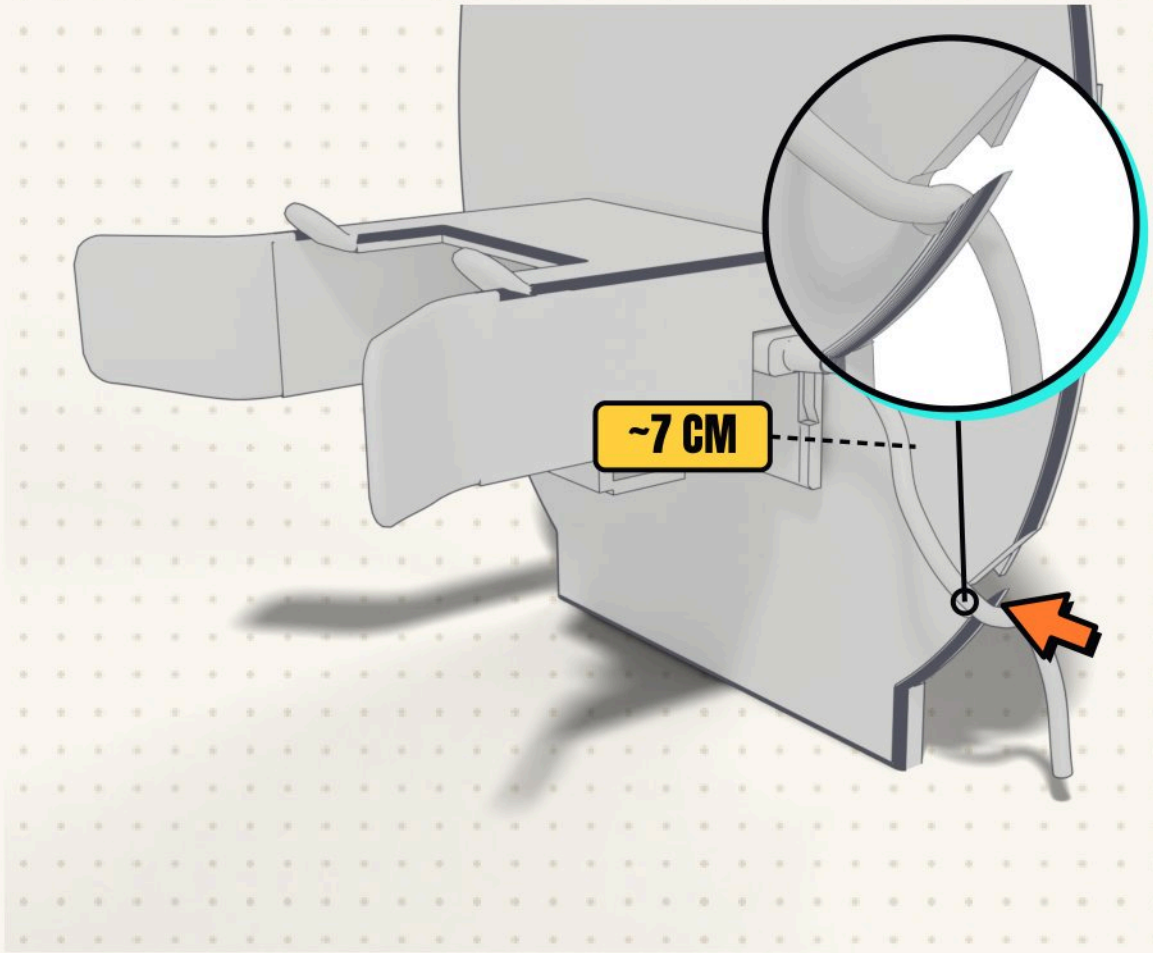
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From behind, push the second pin down through the hole in the crossbar, the carriage and the arm. Cut it off flush.

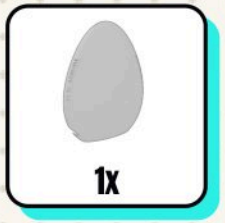
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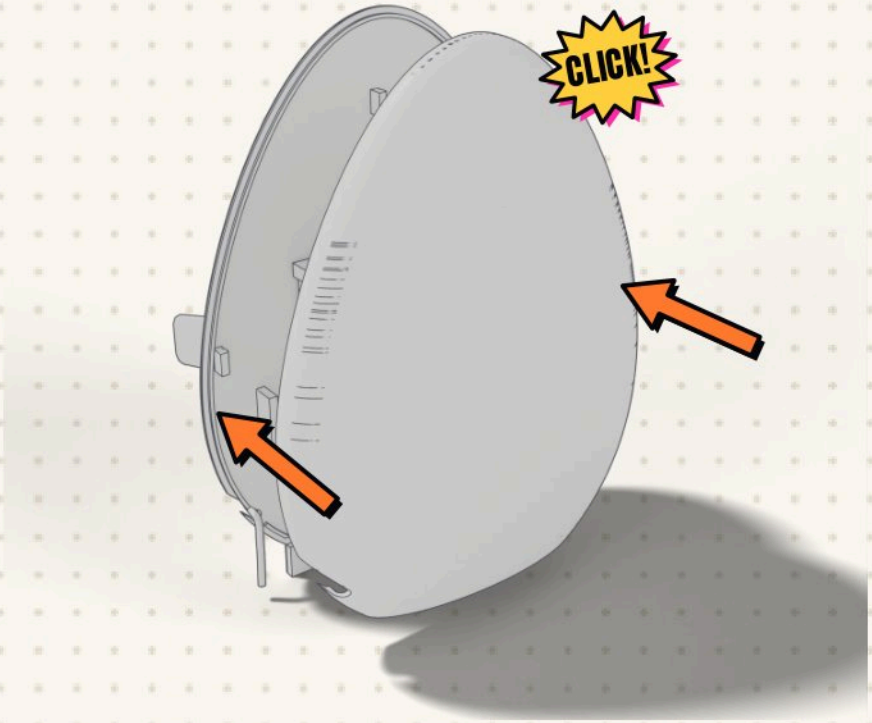


Press the cable into the clip in the egg's edge, near the hood's side. Leave about 7 cm of cable between the hood and the clip, in a loose loop.

The loop lets the hood float: lift the hood and slide it sideways, the cable must never pull on it.

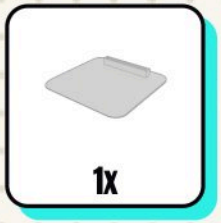
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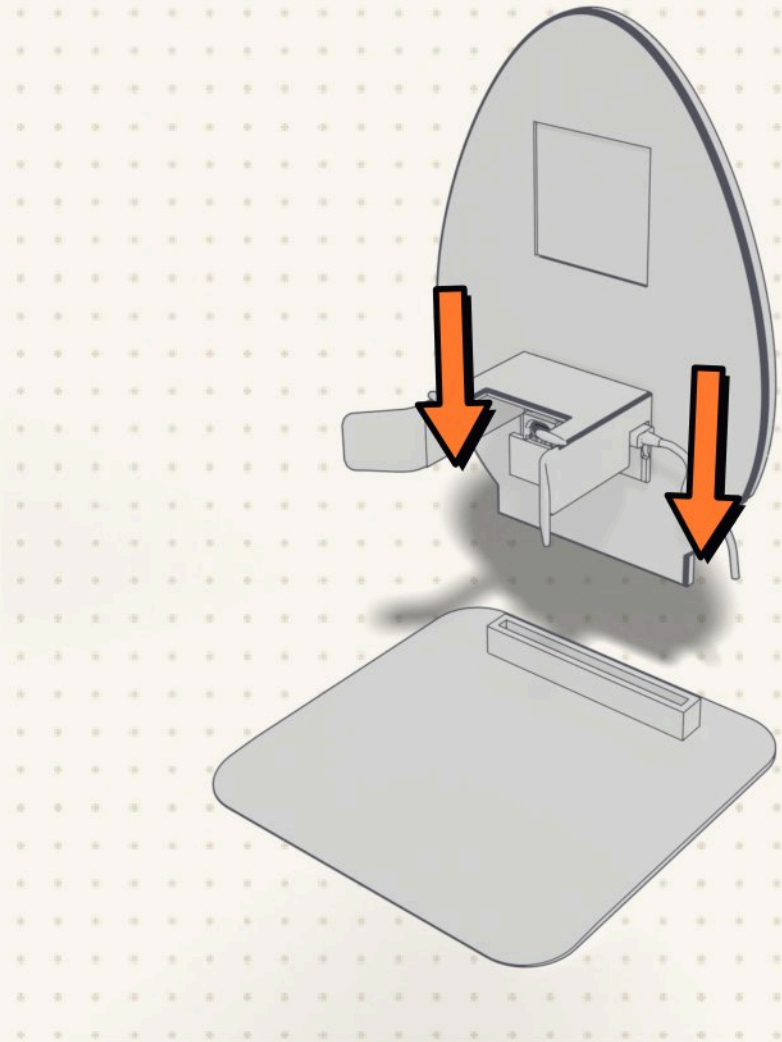




Put the dome on from behind and press it all round until its five snaps click. The cable runs out through its notch at the clip.

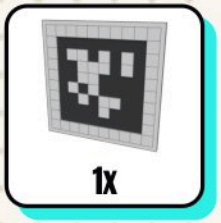
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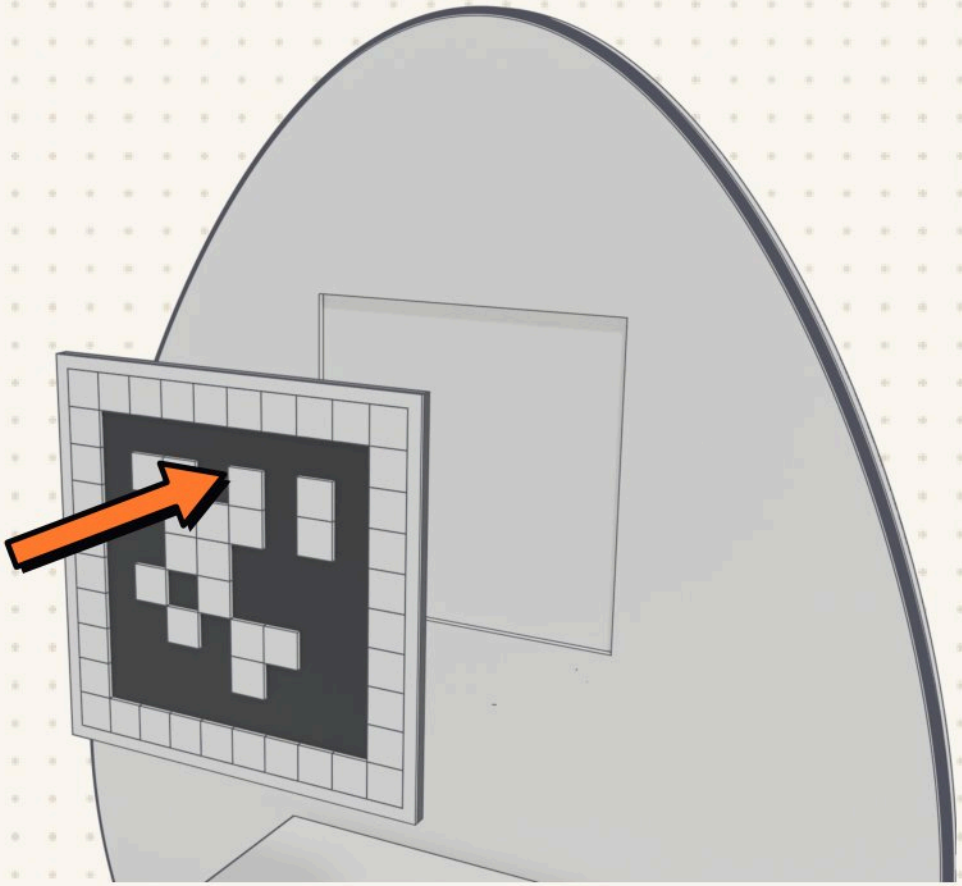




Push the egg down into the base's socket, the hood facing out.

11



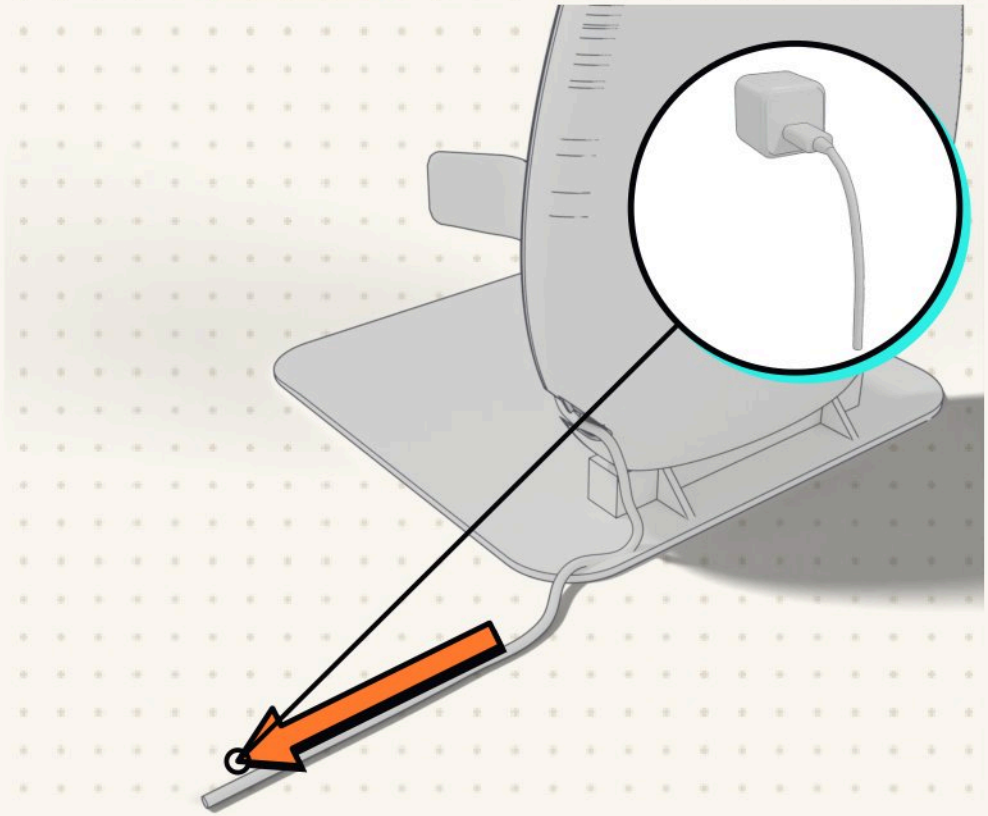


Press the tag into its pocket on the front plate, white cells out.

12

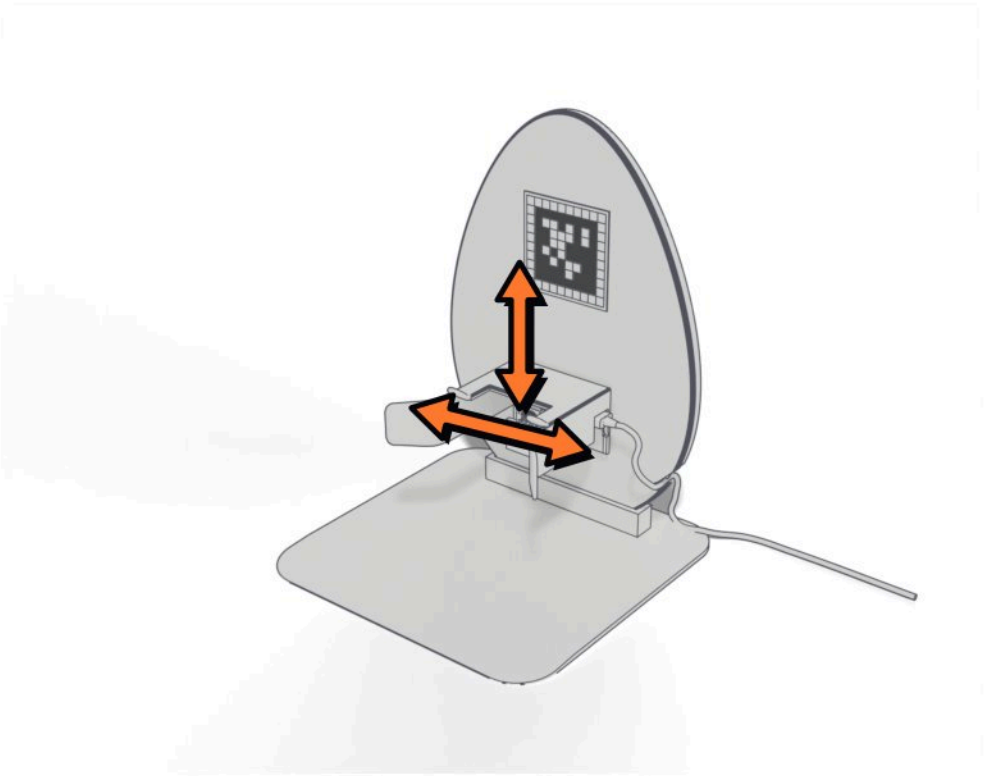


1x



Run the cable along the floor and plug it into the USB-C charger.

CHECK



The hood floats: lift it a little and slide it sideways. It springs back to the middle and down onto its stop, cable and all. That's how it finds Microduck's back.

13



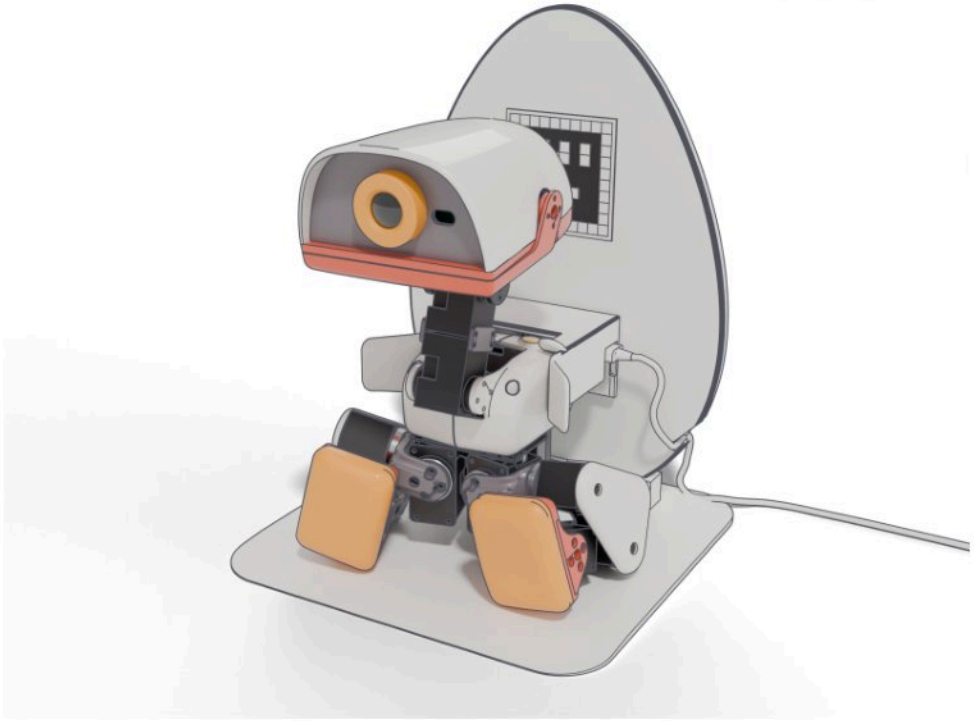
1x



On Microduck: push the magnetic tip into the USB-C port on its back. It stays there.

DONE!

CHARGE!



Stand the dock against a wall and stick the base to the floor with removable mounting tape: Microduck pushes the dock when it backs in and pulls it when it leaves.

Microduck spots the tag, walks up, sits down and scoots back under the hood. The magnets click and USB power comes on. Charged, it scoots out, stands up and walks off.

DOCK, CHARGE, GO



1 BACKS IN



2 CLICK: CHARGING



3 DOZES



4 WAKES UP



5 SCOOTS OUT

GOOD TO KNOW

- **Check your Microduck first.** Recent units have the USB-C port on the back, just above the battery. The hood is shaped for that.
- **Try the adapters first:** plate 7 prints the hood's middle in 33 min. The magnetic adapter, the angle adapter and the duck's tip should all go in as in steps 3-5.
- **Keep the dock in place:** against a wall, the base taped down. Docking pushes it back by a few newtons and leaving pulls it forward; on its own it would slide on a smooth floor.
- **The hood must move freely.** If it rubs, check that the dome's snaps are all in and that the cable's loop isn't tight.
- **Taking the hood off:** take the dome off (pull it straight back), pull the arm's pin out with pliers and pull the hood off.
- **Other cables:** the hood fits this cable's plug and the goobay adapter, the clip a cable up to 5 mm. Both are parameters in `source/flexdock_c.py`.
- **Prototype.** The dock is printed and tried by hand. The docking moves run in simulation so far; Microduck's own software doesn't have them yet.

HAPPY DOCKING!

NT ★ PIN ★ CLICK ★ CLIP ★ SNAP ★ DOCK ★ DOZE ★ GO ★ PRINT ★ PIN ★ CLICK ★ CLIP ★ SNAP ★ DOCK ★ DOZE ★ GO ★